

Zeenath Ara Syed

Forward Deployed Engineer

Dublin, Ireland | s.zeenath.ara@gmail.com | +353 872 352 726 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Eight years of software engineering at Kongsberg Digital delivering production systems inside enterprise client environments. Spent 11 months embedded on-site at Shell's Energy Transition Campus in Amsterdam building an LLM-powered Asset Copilot with Microsoft and presenting a Digital Twin proof-of-concept to prospective customers. Completed an MSc in Robotics and Embedded AI at Maynooth University in August 2026 with a thesis on robotic patient simulators for vestibular clinical training, which received first prize at presentation.

PROFESSIONAL EXPERIENCE

Kongsberg Digital Software Development Engineer L1

Apr 2022 - Aug 2025

Client-Embedded Delivery AI Platform LLM Integration

- Spent 11 months embedded on-site at Shell's Energy Transition Campus, Amsterdam (November 2022 to November 2023). Designed and presented a Digital Twin proof-of-concept to prospective customers, and built KogniTwin's Asset Copilot in partnership with Microsoft using a retrieval-augmented generation pipeline combining GPT-4 and Llama to ground responses in live asset data.
- Designed and shipped a production evaluation framework for the Asset Copilot scoring LLM responses on accuracy, precision, F1 score, and hallucination rate. Cut manual review time by 50% and raised response accuracy by 40% across more than 50 test queries, becoming the team's adopted quality benchmark.
- Led the Energy Nomination Dashboard build in Angular, InfluxDB, and Node.js for 24/7 field workflows, achieving under 100ms latency and a 25% increase in operator engagement.

Kongsberg Digital Software Development Engineer L2 / L3

Jul 2020 - Mar 2022

Dashboard Delivery 3D Visualization Client Portfolio

- Built 3D visualization views in Angular and TypeScript using coordinate-based geometric transforms to render live asset data on the Digital Twin platform for global enterprise clients. Built POC dashboards that directly supported POC-to-contract conversions for the sales team.
- Migrated customer data to MongoDB, built reusable dashboard templates, and introduced Docker and Kubernetes deployment workflows, cutting new-dashboard setup time by 50%.

Kongsberg Digital Software Engineer (Trainee)

Jul 2018 - Jun 2020

Sole Client Ownership Production Delivery SAT

- Took sole ownership of the ConEdison real-time gas-pipeline dashboard for Manhattan. Engineered coordinate-based geometric calculations to render pipelines, valves, and flanges from real-world location data and delivered four monthly production releases on SAT calls with zero critical defects.

BizRuntime IT Services Software Engineer

Jan 2017 - Mar 2018

Full-Stack Applied ML

- Built production features and early neural network models in a full-time entry-level engineering role.

ROBOTICS AND APPLIED ML PROJECTS (MSC, MAYNOOTH UNIVERSITY)

Robotic Eye Movement Simulator for Dix-Hallpike Test

Industrial partnership with St. James's Hospital, Dublin

- Built a physical robotic patient model for Dix-Hallpike vestibular training. The system uses a tendon-driven dual-eye mechanism on Arduino to replicate BPPV eye movements, tracks the clinician's head position via IMU with a quaternion-based pipeline that avoids gimbal lock, and mirrors state in a Unity 3D digital twin at 100Hz. A Google Cloud speech pipeline delivers procedural guidance and evaluates spoken diagnoses via keyword matching.
- Validated results: 27.9°/s torsional slow-phase velocity (published clinical range 15.7 to 38.9), 1.40° IMU mean error, 8.63ms system latency (58ms under target), and 10/10 correct side detection and diagnosis evaluation across all trials.

Probabilistic Intent Recognition for Human-Robot Collaboration

- Proactive manipulation framework on Franka Panda (MuJoCo). Used MediaPipe hand keypoints as input to Bayesian intent inference over a discrete action prior, then minimum-jerk trajectory generation to pre-position the arm before any verbal request. Achieved 93% intent recognition accuracy at 127ms response latency.

EDUCATION

MSc, Robotics and Embedded AI, Maynooth University

2025 - 2026

B.E, Electronics and Communication Engineering, Reva Institute of Technology

2013 - 2017

CERTIFICATIONS

- Fundamentals of Deep Learning, NVIDIA (2026)
- Professional Scrum Master I (PSM I), Scrum.org (2024)

- Applied Computer Vision: Object Detection and Recognition, Udemy (2025)

ACHIEVEMENTS

- **First Prize, MSc Thesis Presentation, Maynooth University (August 2026):** Awarded for the Robotic Eye Movement Simulator for Dix-Hallpike Test, built in partnership with St. James's Hospital. All five technical objectives independently validated against published clinical benchmarks.
- **MUSE Award, Platinum Level, Maynooth University (2026):** The university's highest recognition for student leadership and community engagement.
- **First Place, GenAI Hackathon (2024):** Gas lift optimizer that increased client profit margins by 20%.
- **Ace Award and Spotlight Awards, Kongsberg Digital (2020, 2022):** Recognized for technical leadership, mentoring, and impactful delivery on client-critical products.

VOLUNTEERING AND LEADERSHIP

- Women Techmakers Ambassador, Google Developer Groups (2023 to present)
- President, Toastmasters Club, Kongsberg Digital (2021 to 2022)
- Mentor and Interviewer, Kongsberg Digital (2021 to 2025)